

ABSTRACT

Park Ease – IOT Based Smart Parking Management System

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Abstract

The rapid growth in the number of vehicles has significantly increased the demand for efficient parking management systems. Conventional parking methods often result in traffic congestion, longer waiting times, inefficient space utilization, and unnecessary fuel consumption due to the difficulty in locating vacant parking slots. To address these challenges, **Park Ease – IOT Based Smart Parking Management System** is proposed as an Internet of Things (IoT)-based solution that automates parking management through real-time monitoring and intelligent reservation.

The proposed system utilizes an **ESP32 microcontroller** integrated with **ultrasonic sensors** to detect the occupancy status of individual parking slots. The sensor data is transmitted through Wi-Fi to a centralized **Python Flask** server, where it is processed and stored in a **MySQL** database. A responsive web application developed using **HTML, CSS, JavaScript, and Bootstrap** enable users to register, view real-time parking availability, reserve parking slots during designated reservation periods, and access their parking history. The administrator dashboard provides facilities to monitor slot occupancy, manage reservations, track vehicle entry and exit, and generate parking utilization reports.

A key feature of the system is the **Hybrid Reservation Model**, in which selected parking slots or peak hours are allocated for advance reservations while the remaining slots remain available for walk-in users. This approach improves parking space utilization, minimizes revenue loss due to unused reserved slots, and offers greater flexibility for both users and parking operators.

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